

数据结构习题串讲

「2022 青苗讲堂」

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2022.12.14 18:45-20:45

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习题全程会在平板上板书

板书之后的PDF会在QQ群中发给大家

其他的资料也都会发在QQ群中~

大纲

Ch1 - 基础概念	(时间复杂度分析)
Ch2 - 数组	(KMP)
Ch3 - 栈与队列	(栈、循环队列)
Ch4 - 链表	(循环链表、双向链表)
Ch5 - 树	(基本性质、树的遍历、线索二叉树、森林、胜者树与败者树、堆)
Ch6 - 图	(基本性质、图的遍历、最小生成树、最短路径、拓扑排序)
Ch7 - 排序	(选择、插入、快排、归并、堆排、外排序)
Ch8 - 哈希	(线性探测、平均查找时间)
Ch9 - 二叉排序树	(手撕AVL树)
Ch10- 多路查找树	(B树与B+树基本性质、异同点、手撕B树)

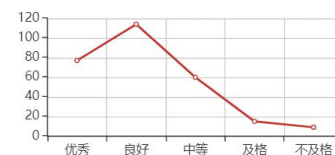
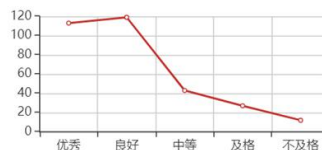
题型 (20级)

选择题 (10*2分)
简答题 (6*10分)
程序设计 (2*10分) [三选二]

课堂成绩分布



课堂成绩分布



基础概念

1. 分析下列程序段的时间复杂度

```
count = 0;  
for(int k = 1 ; k <= n ; k *= 2)  $\log_2 n$   
    for(int j = 1 ; j <= n ; j++)  $n$   
        count++;
```

$O(n \log n)$

2. 分析下列函数的时间复杂度

```
int func(int n)  
{  
    int i = 0, sum = 0;  
    while(sum < n) sum += ++i;  
    return i;  
}
```

$O(\sqrt{n})$

基础概念

3. [21年真题]分析下列程序段的时间复杂度

$d = 1, x = 0;$

$\text{while}(x < n)$

{

$d++;$

$x = x + d;$

}

$x \neq 0$

$x = 0 + 2$

$x = 0 + 2 + 3$

$x = 0 + 2 + 3 + 4$

$x = 1 + 2 + 3 + \dots + k - 1$

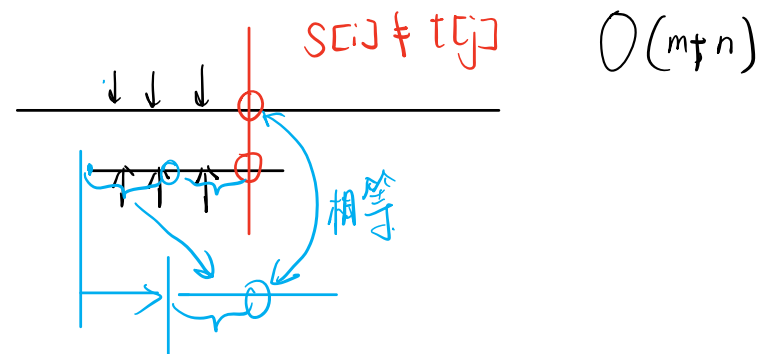
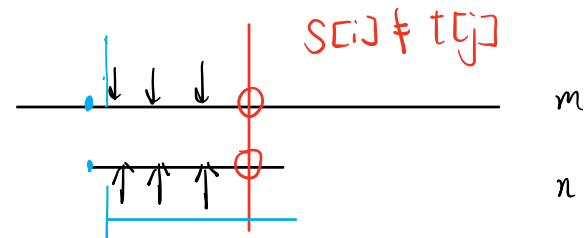
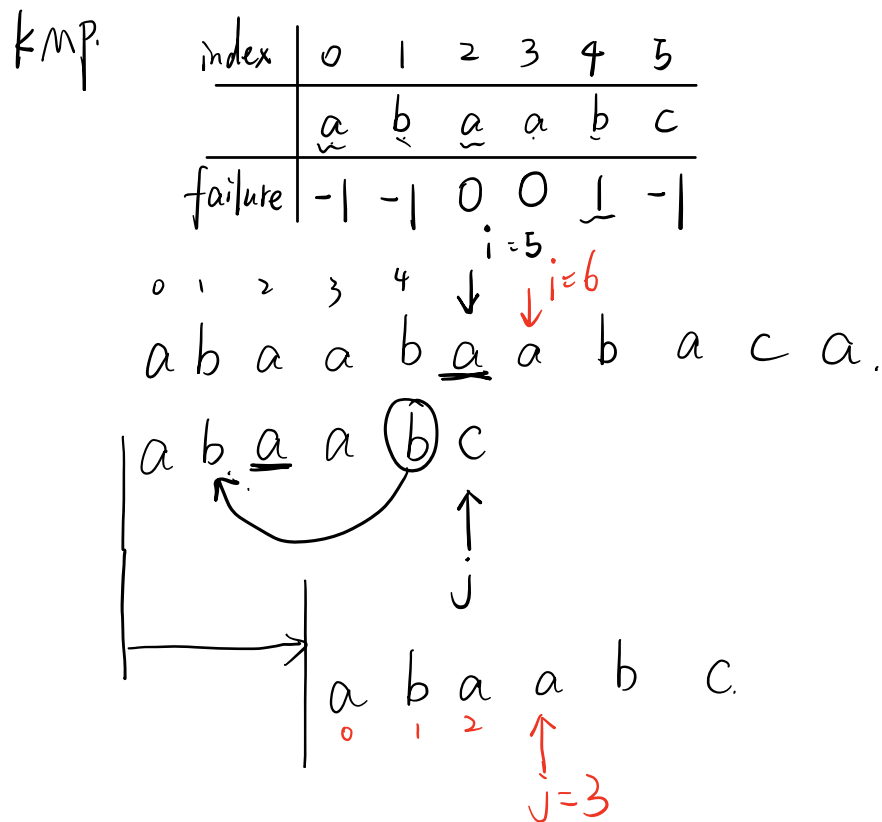
$\rightarrow \frac{(1+k)k}{2} - 1 \rightarrow k^2$

$k^2 = n \quad k = \sqrt{n}$

$O(\sqrt{n})$

数组 (KMP算法的应用)

4. 已知字符串S为“abaabaabacacaabaabcc”，模式串t为“abaabc”。采用KMP算法进行匹配，第一次出现失配($S[i] \neq t[j]$)时， $i=j=5$ ，则下次开始匹配时， i 和 j 的值分别为6，3。



数组 (KMP算法的应用)

5. 已知字符串T为“abaabaabcabaabc”，模式串t为“abaabc”。采用KMP算法进行匹配，到匹配成功为止，在匹配过程中进行的单个字符间的比较次数为9。

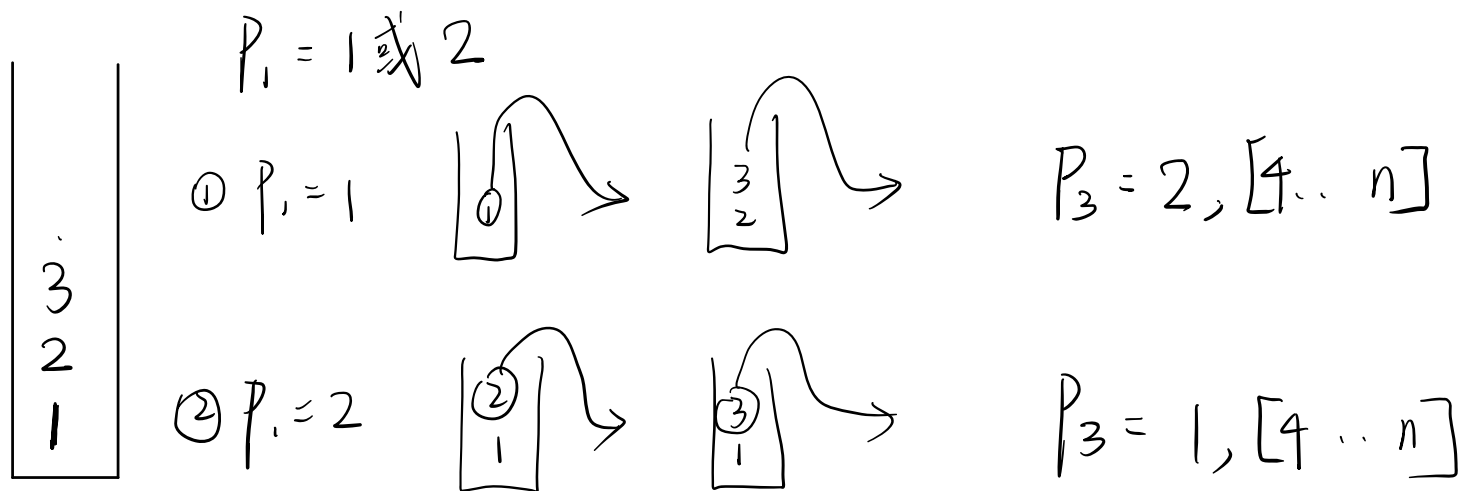
index	0	1	2	3	4	5
	<u>a</u>	<u>b</u>	<u>a</u>	a	b	c
failure	-1	-1	0	0	1	-1

↓ ↓
a b a a b a a b c a b a a b c
↗
| a b a a b c
|
→ | a b a a b c
↑

$$\begin{aligned} &+6 \\ &+3 \\ &=9 \end{aligned}$$

栈与队列

6. 一个栈的入栈序列为 $1, 2, 3, \dots, n$, 出栈序列为 $P_1, P_2, P_3, \dots, P_n$, 若 $\underline{P_2=3}$, 则 P_3 可能取值的个数为 $\underline{n-1}$



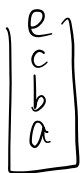
除33的元素

$n-1$

栈与队列

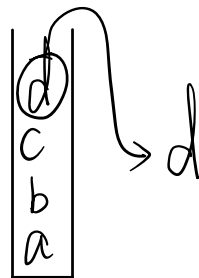
7. 元素a,b,c,d,e,依次进入初始为空的栈中, 若元素进栈后可以停留, 可以出栈, 直到所有元素都出栈, 则在所有可能的出栈序列中, 以元素d开头的序列个数是 4.

1. 先进e

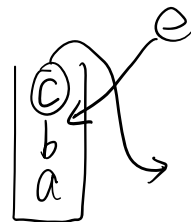


e c b a

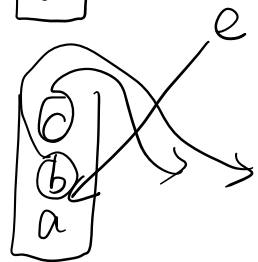
2. 先出



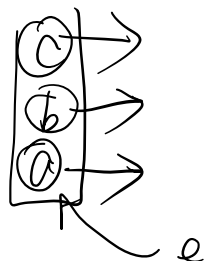
d



c e b a



c b e a



c b a e.

栈与队列

8. 若元素a,b,c,d,e,f依次进栈，允许出栈、退栈操作交替进行，但不允许连续进行三次退栈操作，则不可能得到的出栈序列是 D

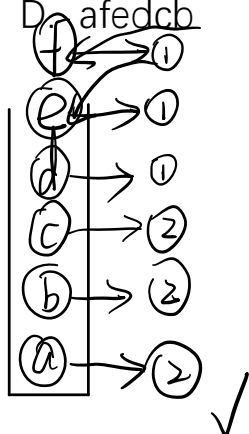
A. dcebfa

B. cbdaef

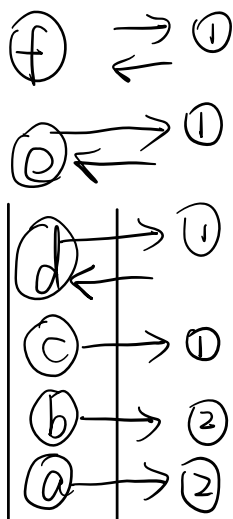
C. bcaefd

D. afedcb

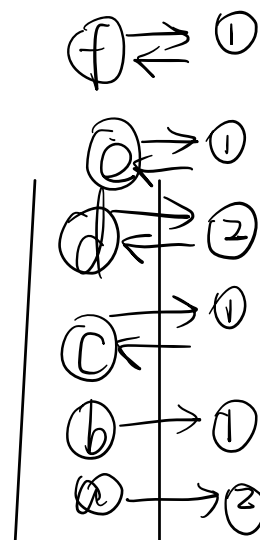
A



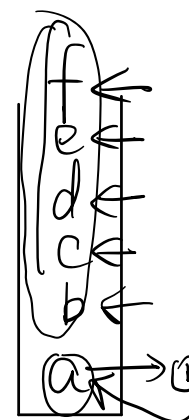
B



C

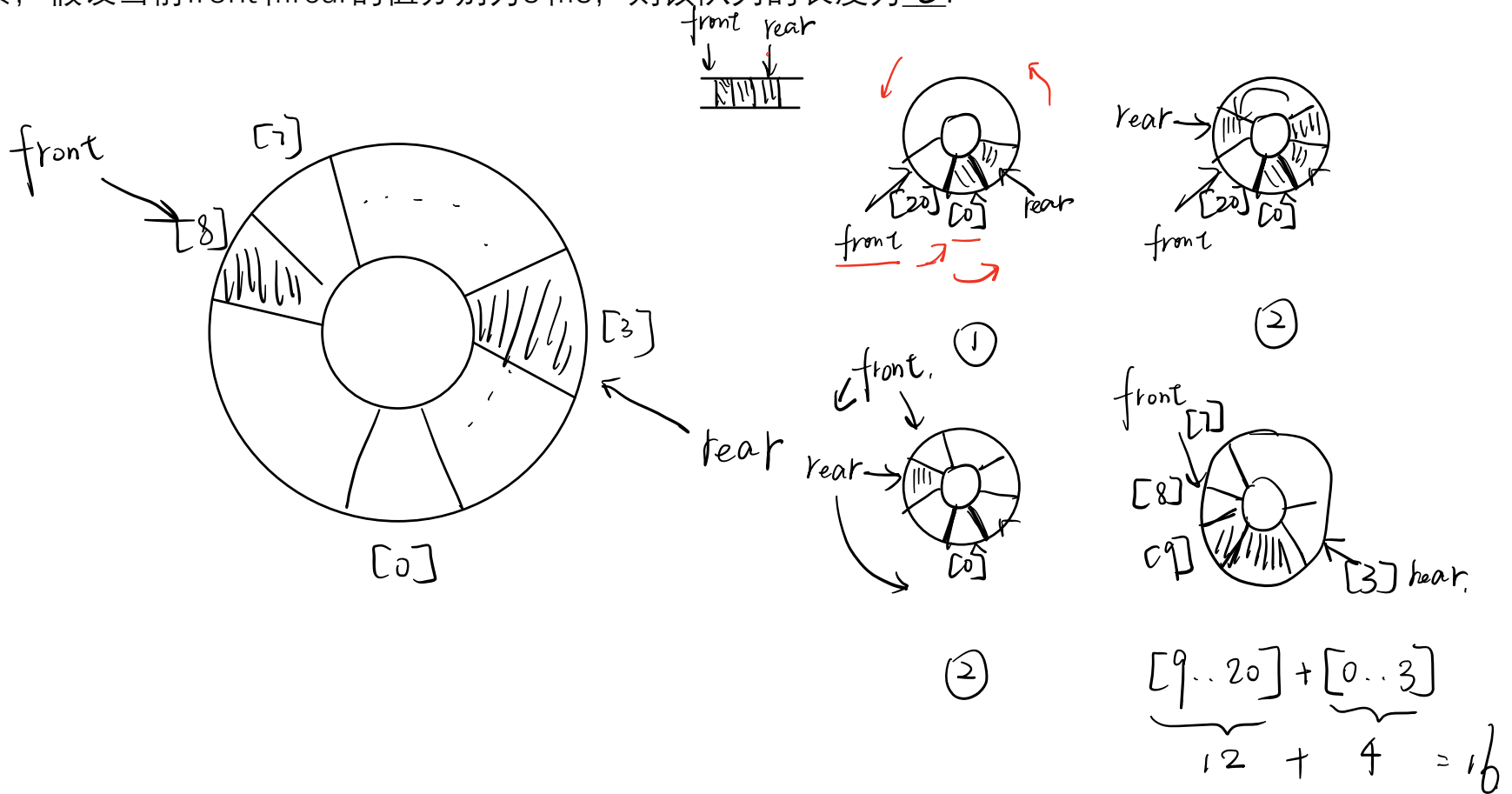


D



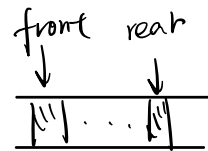
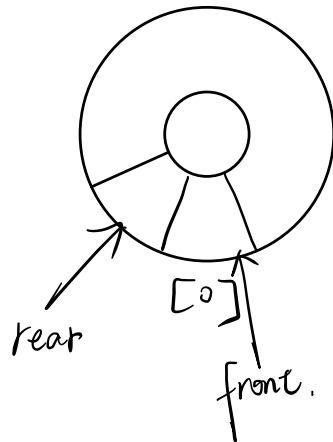
栈与队列

9. 已知循环队列的存储空间为A[21], front指向队头元素的前一个位置, rear指向队尾元素, 假设当前front和rear的值分别为8和3, 则该队列的长度为16.



栈与队列

10. 已知循环队列存储在一维数组 $A[0 \cdots n-1]$ 中，且队列非空时front和rear分别指向队头和队尾元素，若初始时队列为空，且要求第一个进入队列的元素存储在 $A[0]$ 处，则初始时front和rear的值分别是___。



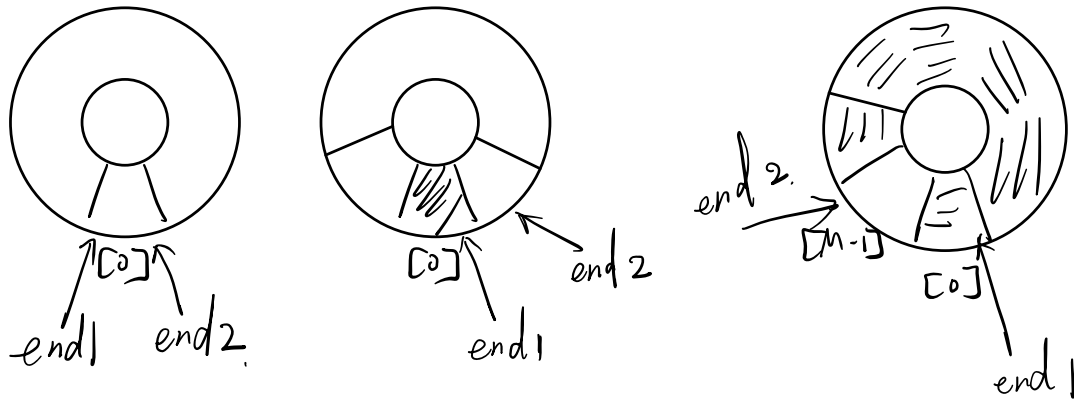
rear += 1
 $A[\text{rear}] = \text{num}$

front = 0
rear = n - 1

栈与队列

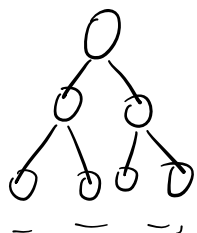
11. 已知循环队列存储在一维数组 $A[0 \cdots M-1]$ 中， $end1$ 指向队头元素， $end2$ 指向队尾的下一个位置，假设队列两端均可以进行入队和出队操作，队列中最多能容纳 $M-1$ 个元素，初始时空，系列判断队空和队满的条件，正确的是(A) [A]

- A. 队空: $end1 == end2$; 队满: $end1 == (end2 + 1) \bmod M$ ✓
B. 队空: $end1 == end2$; 队满: $end2 == (end1 + 1) \bmod (M - 1)$
C. 队空: $end2 == (end1 + 1) \bmod M$; 队满: $end1 == (end2 + 1) \bmod M$
D. 队空: $end1 == (end2 + 1) \bmod M$; 队满: $end2 == (end1 + 1) \bmod (M - 1)$



树 (基本性质)

12. 若一颗完全二叉树有768个结点，则该二叉树中叶结点的个数为384。 [384]



- ① $1 = 2^0$
- ② $2 = 2^1$
- ③ $4 = 2^2$

$$1 \ 2 \ 4 \ 8 \ \dots \ 2^{l-1}$$

l层满 = 叉树: $2^0 + 2^1 + \dots + 2^{l-1} = 2^l - 1$

$$\underbrace{1 \ \dots \ 1}_{l-1 \text{ 位}} \ 1 \ 0 = \underbrace{1 \ 0 \ 0 \ \dots \ 0}_{l \text{ 位}} - 1 = 2^l - 1$$

$$2^{10} = 1024$$

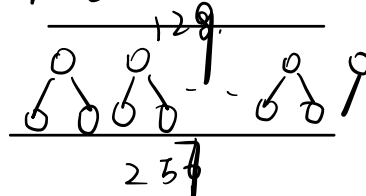
$$2^9 = 512$$

9层满 = 叉树 511 结点

第9层 $2^8 = 256$ 结点



$$768 - 511 = 257$$



$$257 + 256 - 129 = 384$$

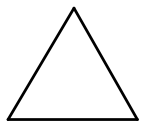
树 (基本性质)

13. 在一颗完全二叉树中, 其根序号为1, (A) 可以判定序号为p和q的两个结点是否在同一层。[A]

- A. $\lfloor \log p \rfloor = \lfloor \log q \rfloor$
- B. $\log p = \log q$
- C. $\lfloor \log p \rfloor + 1 = \lfloor \log q \rfloor$
- D. $\lfloor \log p \rfloor = \lfloor \log q \rfloor + 1$

层序号

[1层最大序号+1, 1层最大序号]



$$2^{l-1} - 1 + 1$$

$$= 2^{l-1}$$

$$\downarrow \log_2$$

$$l-1$$

$$2^l - 1$$

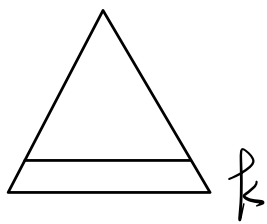
$$\lfloor \log_2 \rfloor$$

$$l-1$$

=

树 (遍历)

14. 设一颗非空完全二叉树T的所有叶结点均位于同一层，且每个非叶结点都有两个子节点。若T有k个叶结点，则T的结点总数为___。[2^{log₂ k} - 1]



$$2^{l-1} = k$$

$$l = \log_2 k + 1$$

$$2^l - 1 = 2^{\log_2 k + 1} - 1 = 2^{\log_2 k} \cdot 2 - 1 = 2k - 1$$

度数=2 点 数量 n_2

叶结点 数量 n_0

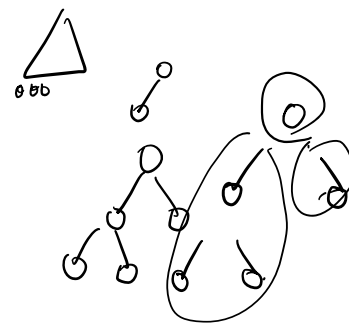
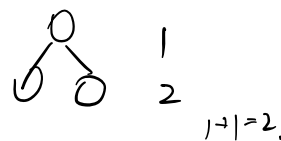
$$n_0 = n_2 + 1$$

$$N = n_0 + n_1 + n_2 \quad ①$$

$$N - 1 = n_1 + 2n_2 \quad ②$$

$$-1 = -n_0 + n_2$$

$$n_0 = n_2 + 1 \quad \checkmark$$

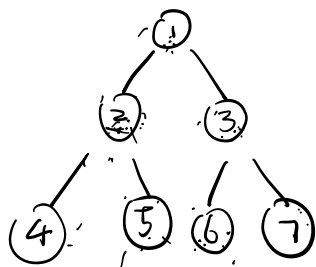
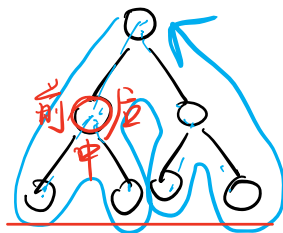


树 (遍历)

visit(current Node) ①
visit(left Child) ②
visit(right Child) ③

15. (true or false) 在二叉树的前序、中序、后序序列中，所有叶子结点的先后顺序都相同

①②③ ②①③ ②③①



前序: 4 5 6 7

中序: 4 5 6 7

后序: 4 5 6 7

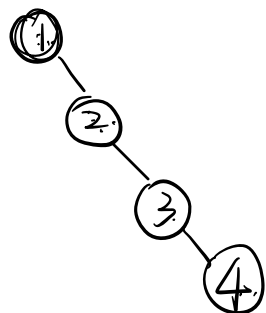
树 (遍历)

16. 若一颗二叉树的前序遍历序列和后序遍历序列分别为1, 2, 3, 4和4, 3, 2, 1 则该二叉树的中序遍历序列不可能是 ()

- A. 1,2,3,4
- B. 2,3,4,1
- C. 3,2,4,1
- D. 4,3,2,1

根结点=1

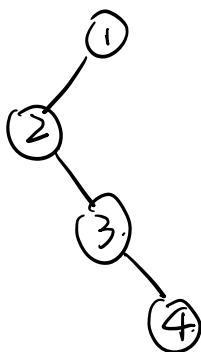
A: 前 1 2 3 4
中 1 2 3 4



后 4 3 2 1 ✓

中=前 left child = null

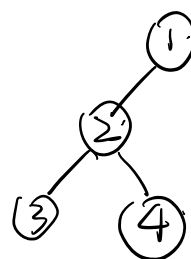
B: 前 1 2 3 4
中 2 3 4 1



后: 4 3 2 1 ✓

left out right

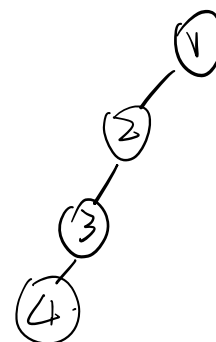
C: 前 1 2 3 4
中 3 2 4 1



后: 3 4 2 1

X

D: 前 1 2 3 4
中 4 3 2 1

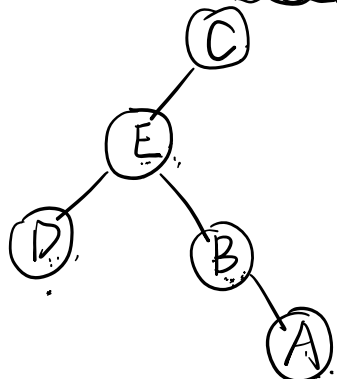


后 4 3 2 1 ✓

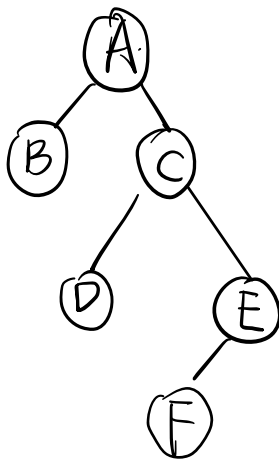
中=后 right child = null

树 (遍历)

17. 若一颗二叉树后序序列为 DABEC 中序序列为 DEBAC, 则先序序列为 CEPBA



18. 若一颗二叉树层序序列为 ABCDEF, 中序序列为 BADC~~FE~~, 则先序序列为 ABCDEF.



树（遍历）

19. 下列序列中，不能唯一确定一颗二叉树的是 

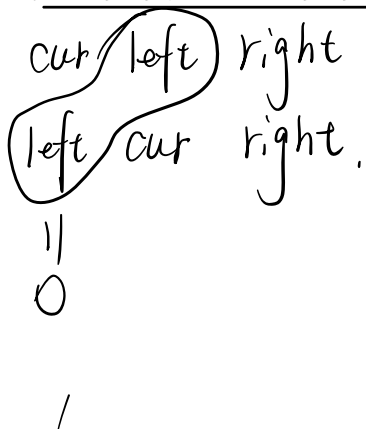
- A. 层序和中序 ✓
- B. 先序和中序 ✓
- C. 后序和中序 ✓
- D. 先序和后序

树 (遍历)

20. 要使一颗非空二叉树的先序序列和中序序列相同, 其所有非叶节点必须满足的条件是

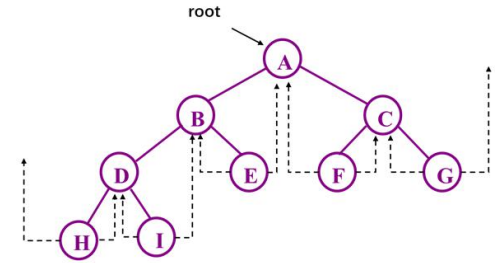
- A. 只有左子树
- B. 只有右子树
- C. 结点的度均为1
- D. 结点的度均为2

A



树 (线索二叉树)

false, 正常: 
true, 叶结点/度=1  作线索用



21. [21年真题]程序填空

①rightChild, leftChild ②leftChild, rightChild

T ThreadedInorderIterator::Next()*

{// Return the inorder successor of currentNode 中序后继结点

// in a threaded binary tree

ThreadedNode<T> *temp = currentNode → rightChild ;

if (!currentNode → rightThread) // false.

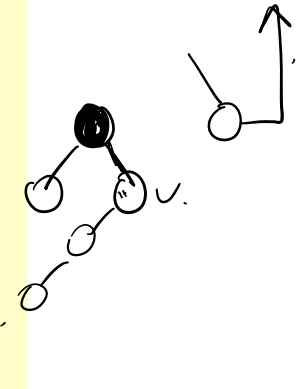
while (!temp → leftThread) temp = temp → leftChild ;

currentNode = temp;

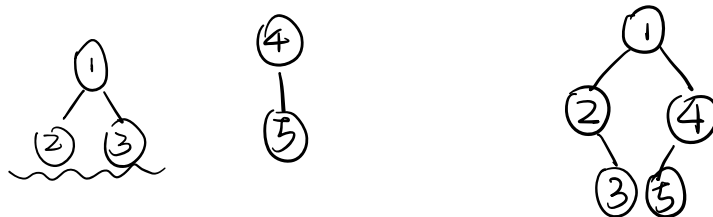
if (currentNode == root) return 0; //no next

else return ¤tNode → data;

}



树 (森林)



22. 将森林转化成对应的二叉树, 若在二叉树中, 结点u是结点v的父节点, 则在原来的森林中, u和v可能具有的关系有_____.

- ①父子关系 ②兄弟关系 ③u的父节点和v的父节点是兄弟关系



树 (森林)



23. 将森林F转化为对应的二叉树T, F中叶结点的个数为 ①

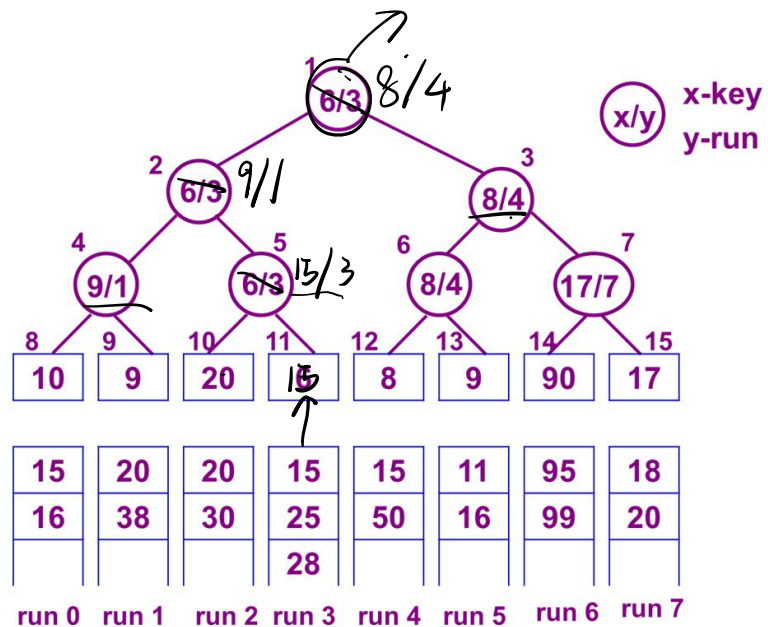
- A. T中叶结点的个数
- B. T中度为1的结点个数
- C. T中左孩子为空的结点个数
- D. T中右孩子为空的结点个数



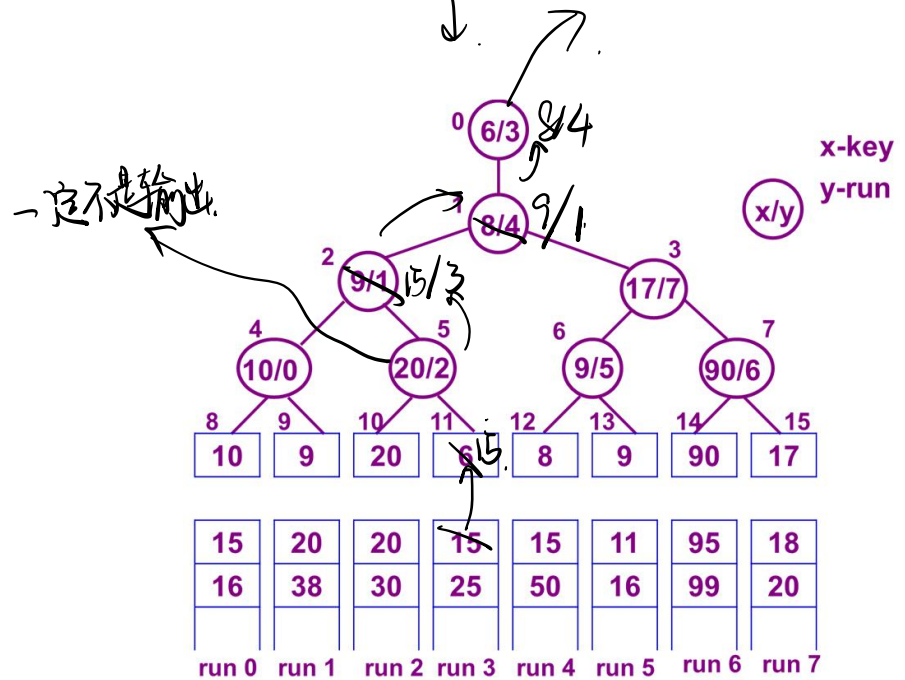
24. 若森林F有15条边, 25个结点, 则F包含的树的个数为 10.

树: n 个结点 $n-1$ 条边

树（胜者树与败者树）

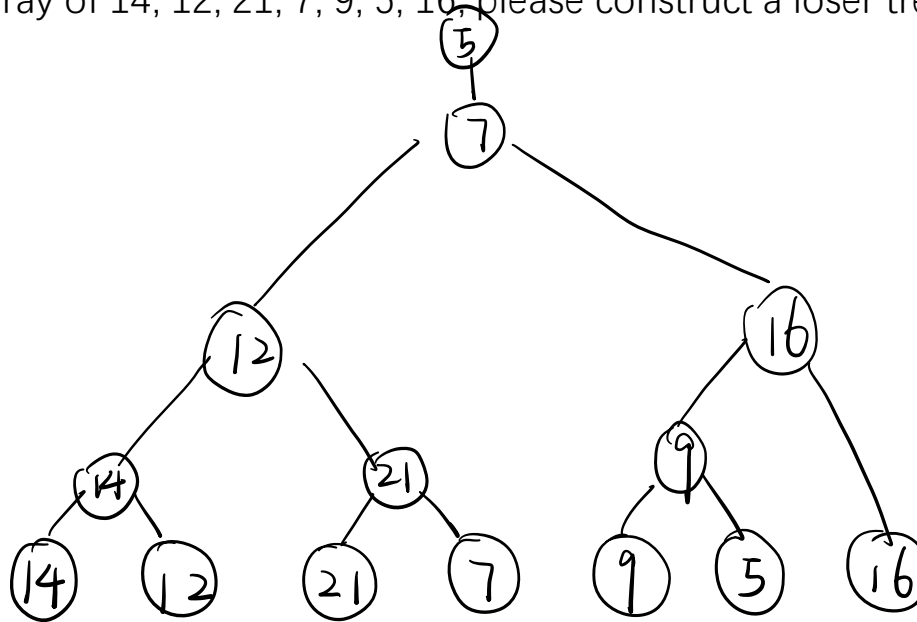


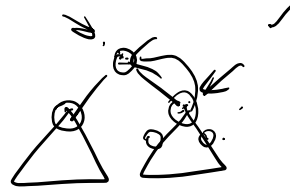
每次与兄弟节点比较。



树（胜者树与败者树）

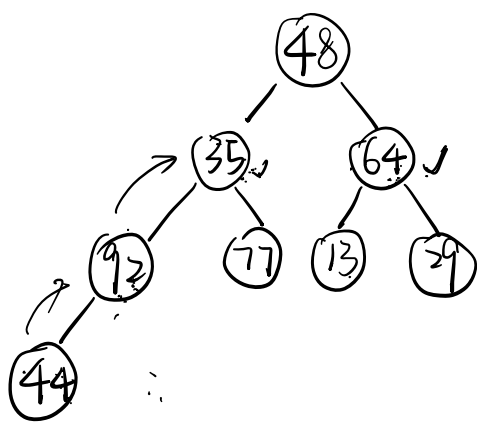
25. Given an array of 14, 12, 21, 7, 9, 5, 16, please construct a loser tree.





树 (堆)

26. Assume a list is {48,35,64,92,77,13,29,44}, firstly insert these items to an empty complete Binary Tree according to the sequence one by one, then please heapify the complete Binary Tree and implement the heap sort. Please draw the whole heapifying process and sorting process.



$$\lfloor \frac{8}{2} \rfloor = 4$$

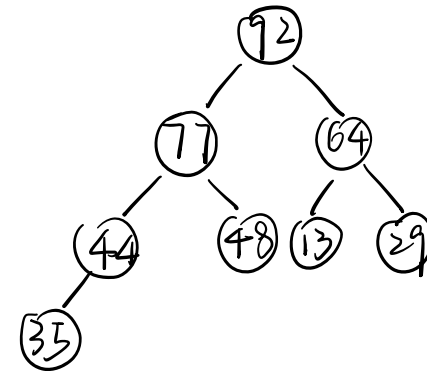
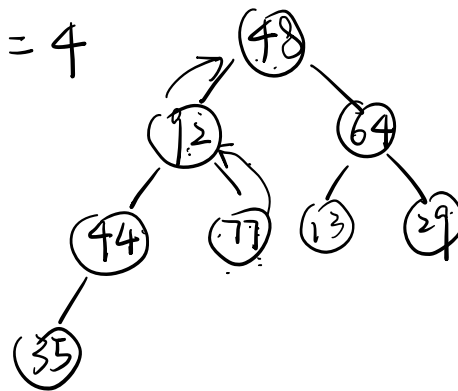
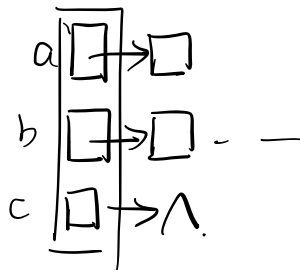


图 (遍历)

	a	b	c
a		1	
b	0		0
c		1	



27.

邻接矩阵表示的图，DFS的时间复杂度？ BFS的时间复杂度？

$O(n^2)$

$O(n^2)$

邻接表表示的图，DFS的时间复杂度？ BFS的时间复杂度？

$O(n+e)$

$O(n+e)$

Visit[] DFS(s)

```

S
visit[s] ← true
for s's every adjacent node m
  if !visit[m]
    DFS(m)
  
```

BFS(s)

```

visit[s] ← true
for s's every adjacent node m
  enqueue(m)
  
```

while (!q.isempty())

visit(m)

for m's every adjacent node x

if !visit(x)

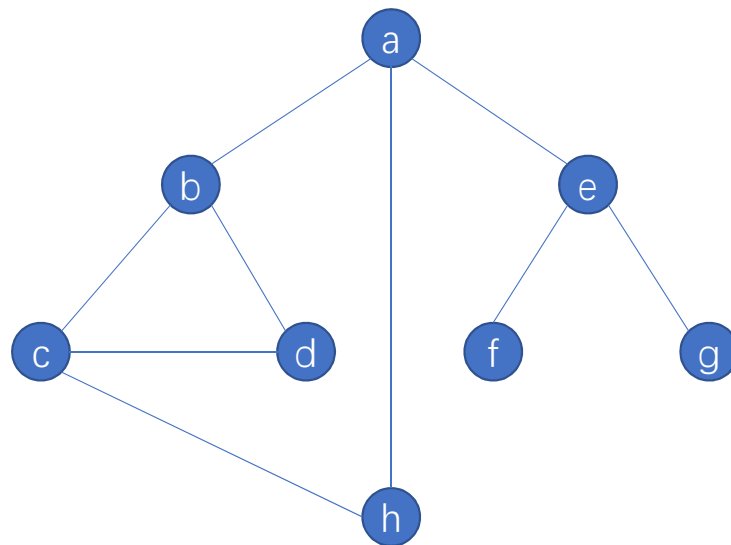
enqueue(x)

m = dequeue()

图 (遍历)

28. 对左边无向图遍历, 不是BFS序列的是()

- A. hcabdegf
- B. eafgbhcd
- C. dbcahefg
- D. abcdhefg



图（遍历）

29. 下列选项中，不是DFS序列的是

- A. 15432
- B. 13254
- C. 12543
- D. 12345

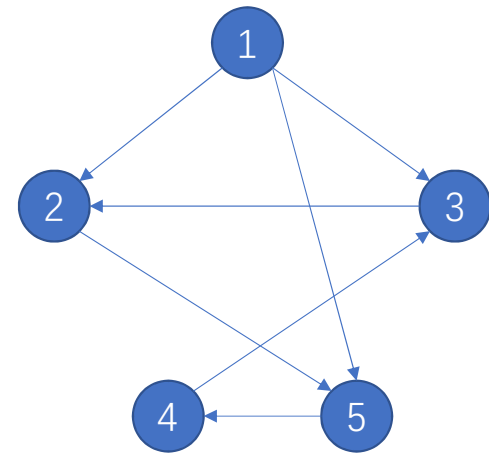
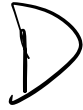


图 (遍历)

30. 设计一个算法，判断一个图G是否为一棵树。

DFS(s, n, e)

n += 1

S

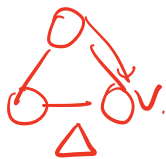
visit[s] ← true.

for s's every adjacent node m

if !visit[m]

e += 1

DFS(m)



if $n - 1 = e$ true

$2 * (n - 1) = e \checkmark \&\& N = n$

① 无环连通图

树:

② 结点数 n 边数 n-1

↑ n 个结点

连通



0

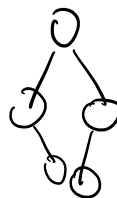


图 (最小生成树)

31. Given a weighted graph $G=(V, E)$ with $V=\{1,2,3,4,5,6,7\}$;

$E=\{(1,2)3,(1,3)5,(1,4)8,(2,5)10,(2,3)6,(3,4)15,(3,5)12,(3,6)9,(4,6)4,(4,7)20,(5,6)18,(6,7)25\}$

, please use Kruskal's Algorithm to build a minimum-cost spanning tree T by adding edges to T one at a time.

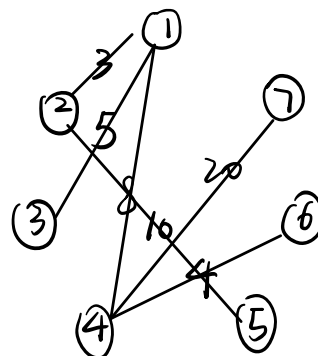
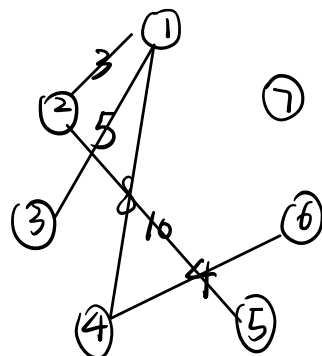
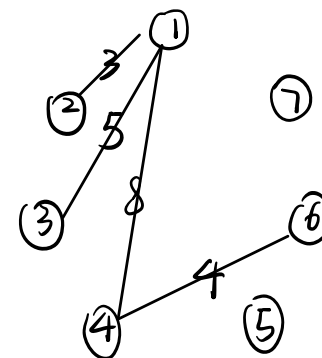
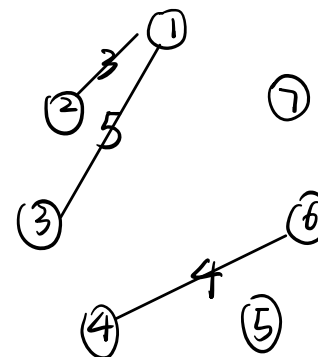
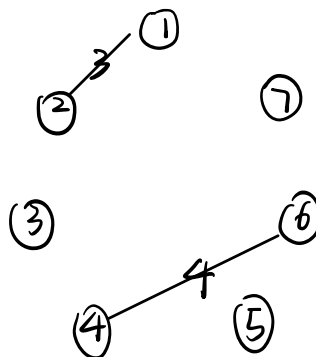
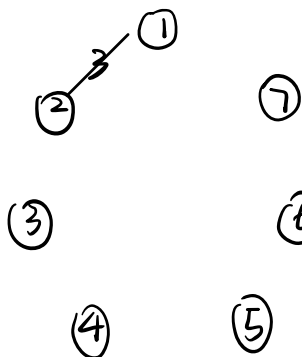
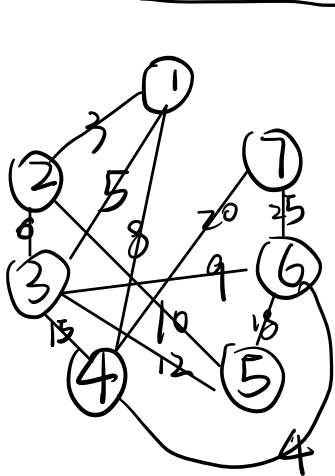


图 (最小生成树)

32. Given a weighted graph $G=(V, E)$ with $V=\{1, 2, 3, 4, 5, 6, 7\}$; $E=\{(1,2) \text{ 3}, (1,3) \text{ 5}, (1,4) \text{ 8}, (2,5) \text{ 10}, (2,3) \text{ 6}, (3,4) \text{ 15}, (3,5) \text{ 12}, (3,6) \text{ 9}, (4,6) \text{ 4}, (4,7) \text{ 20}, (5,6) \text{ 18}, (6,7) \text{ 25}\}$, please use **Prim's Algorithm** to build a minimum-cost spanning tree T by adding edges one at a time.

选1为源点

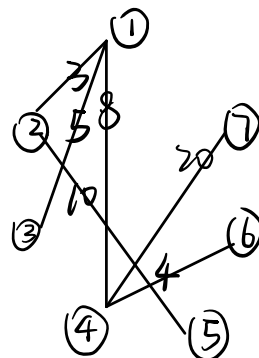
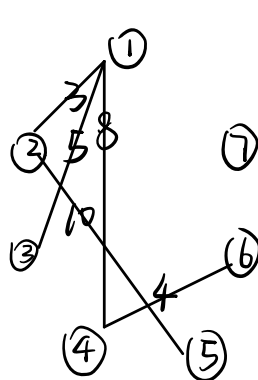
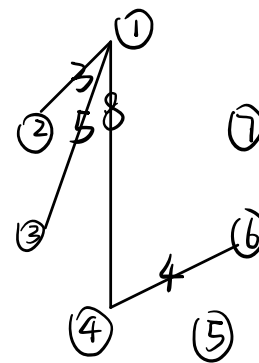
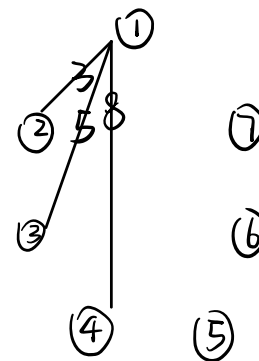
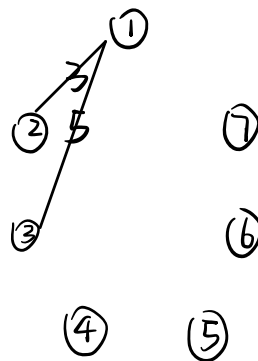
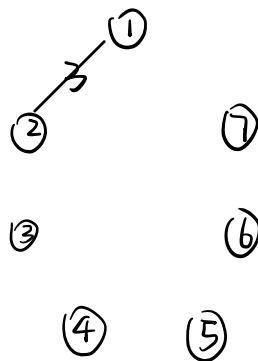
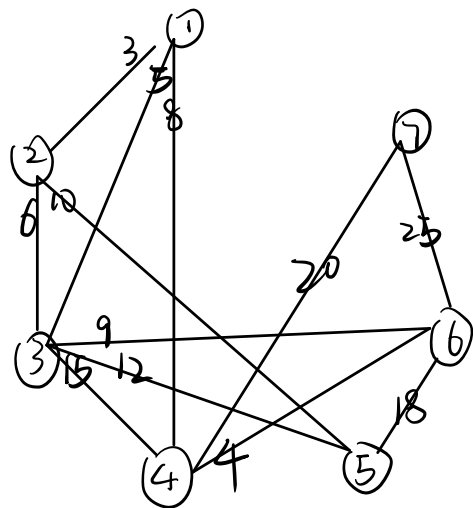


图 (最小生成树)

33. 对最小生成树的描述, 正确的是 (A)
- A. 最小生成树的代价唯一 ✓
 - B. 所有权值最小的边一定会出现在所有的最小生成树中 ✗
 - C. 使用Prim算法从不同顶点开始得到的最小生成树一定相同 ✗
 - D. 使用Prim算法和Kruskal算法得到的最小生成树 总不相同 ✗

图（最短路径）

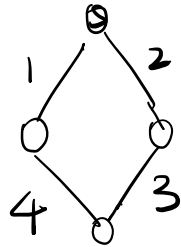
34. [20年真题]试证明Dijkstra算法的合理性。

图 (Dijkstra与Prim)

集合 S
 $\text{dist}[w] ? \text{dist}[u] + \text{length}(u, w).$

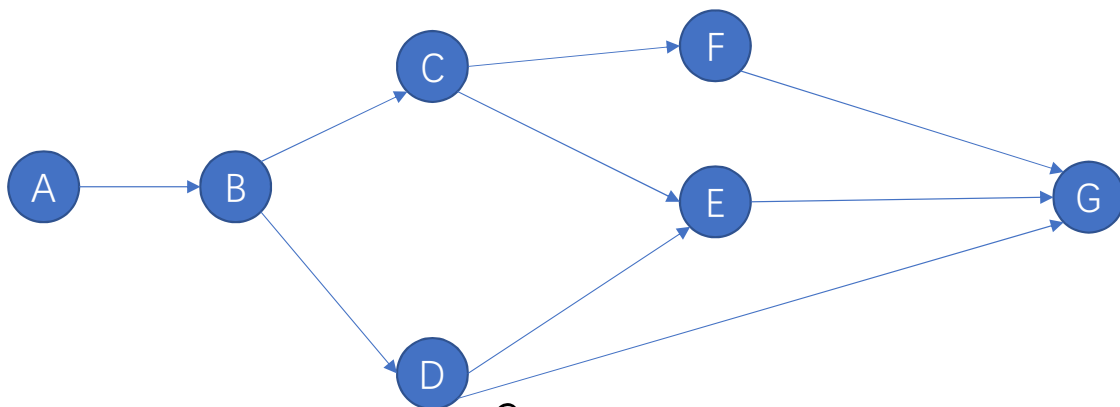
源点 s
 $\text{dist}[w]$ $\text{length}(u, w).$
 环.

35. 一连通无向图，边非负权值，问用Dijkstra算法是否能给出一颗最小生成树，该树是否一定是最小生成树？为什么？



图（拓扑排序）

36. 下图所示有向图的拓扑序列一共有5个.



Handwritten analysis of topological sequences:

From node A, the sequence starts with AB. From node B, there are two choices: C or D.

- If the sequence is ABC, then from C, it can go to F or E. From F, it goes to G. From E, it goes to G. This results in the sequence ABCFG.
- If the sequence is ABD, then from D, it can go to E or G. From E, it goes to G. This results in the sequence ABDEG.

Therefore, the topological sequences are: ABCFG, ABDEG, BACFG, BADEG, and DBCEG.

图

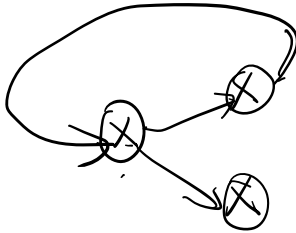
37. 下面的^{①②}可以判断一个图是否有环.

①DFS ②Topological Sort ③Dijkstra



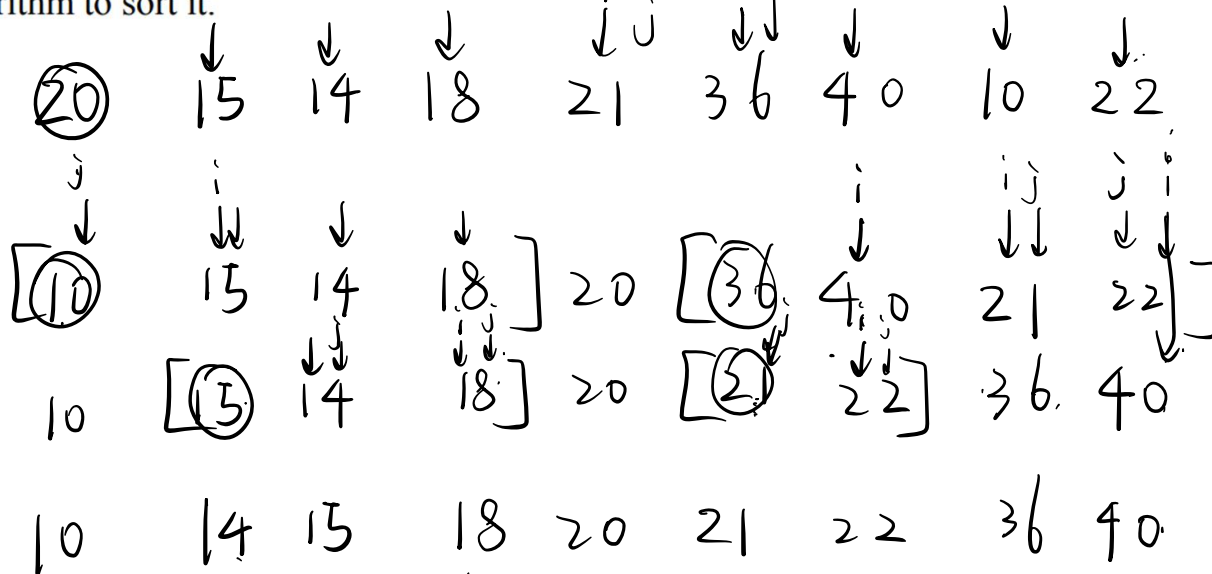
Visit.

前驱节点



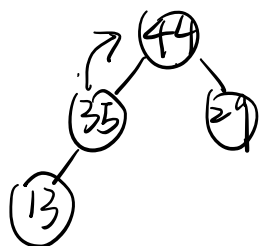
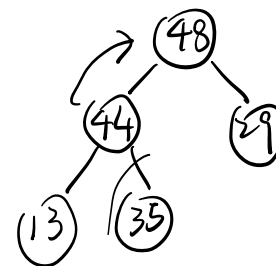
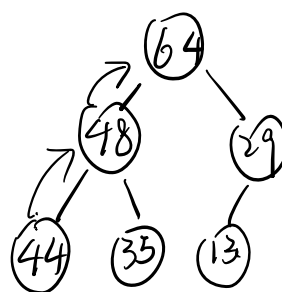
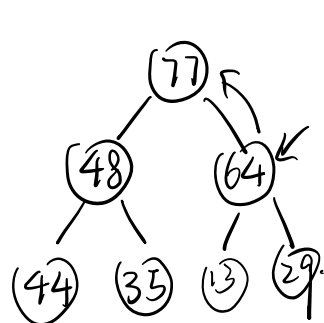
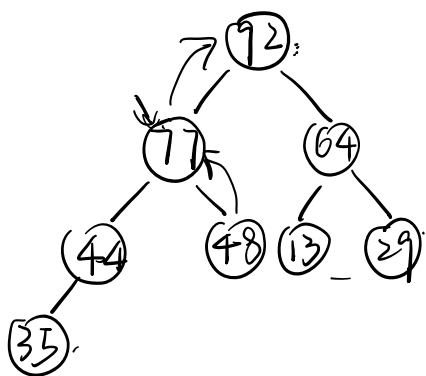
排序

38. Given a list (20, 15, 14, 18, 21, 36, 40, 10, 22), please describe the process that we use **Quick Sort** algorithm to sort it.



排序

39. Assume a list is {48,35,64,92,77,13,29,44}, firstly insert these items to an empty complete Binary Tree according to the sequence one by one, then please heapify the complete Binary Tree and implement the heap sort. Please draw the whole heapifying process and sorting process.



[64, 77, 92]

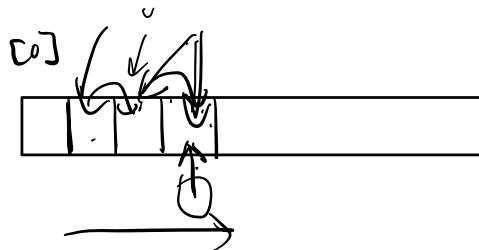
[48, 64, 77, 92]

[44, 48, 64, 77, 92]

[35, 44, 48, 64, 77, 92]

[13, 29, 35, 44, 48, 64, 77, 92]

排序



40. [21年真题] $(10')$

对一个大多数元素已经有序的序列进行内排序，应该使用选择排序还是插入排序，为什么？

$O(n^2)$

$\sim O(n)$

哈希

index	3	7	7	0	9	7	10	9	8
-------	---	---	---	---	---	---	----	---	---

41. 设序列为{47, 0, 29, 11, 9, 84, 54, 20, 30}, 哈希表长13(填充因子 $\alpha = 9/13 = 0.69$), 哈希函数为 $h(key) = key \bmod 11$, 利用Linear Probing解决冲突。

1) 画出哈希表

2) 求平均查找长度ASL

1)

index	0	1	2	3	4	5	6	7	8	9	10	11	12
key	11	30		47				7	29	9	84	54	20

2)

$$ASL = \frac{1 + 1 + 2 + 1 + 1 + 4 + 2 + 4 + 7}{9} = 2.56$$

二叉排序树 (AVL树)

A B C D E F G
H I J K L M N
O P Q R S T U
V W X Y Z

42. Start with an empty AVL tree and perform the following sequence of insertions: DECEMBER, JANUARY, APRIL, MARCH, JULY, AUGUST, OCTOBER, FEBRUARY, NOVEMBER, MAY, JUNE. Use the strategy of *Avl::Insert* to perform each insert. Draw the AVL tree following each insertion and state the rotation type (if any) for each insert.

① Insert DEC
DEC

② Insert JAN
DEC
JAN

③ Insert APR
DEC
APR JAN

④ Insert MARCH
DEC
APR JAN
MARCH

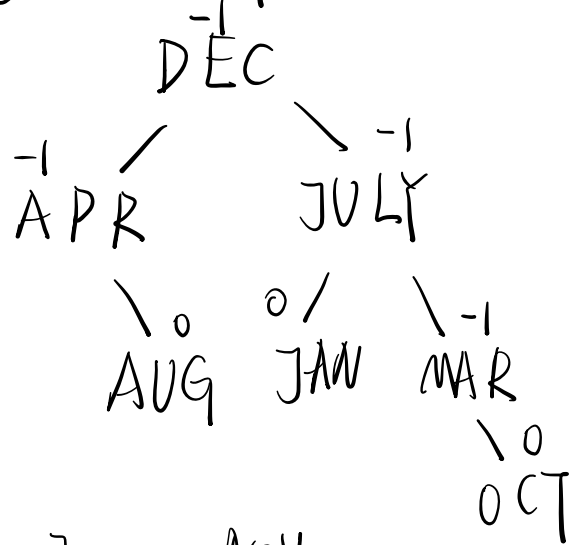
⑤ Insert JULY
DEC
APR JAN
MARCH
JULY

RL
→
DEC
APR JULY
JAN MARCH

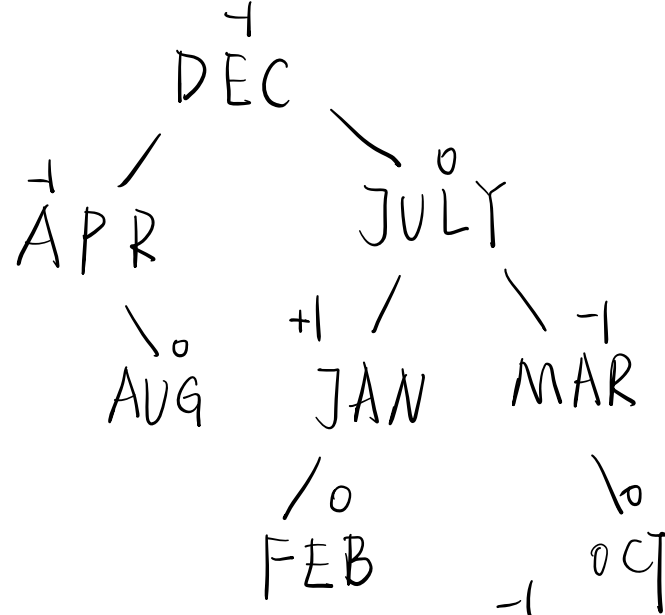
⑥ Insert AUG
DEC
APR JULY
AUG JAN MARCH

二叉排序树 (AVL树)

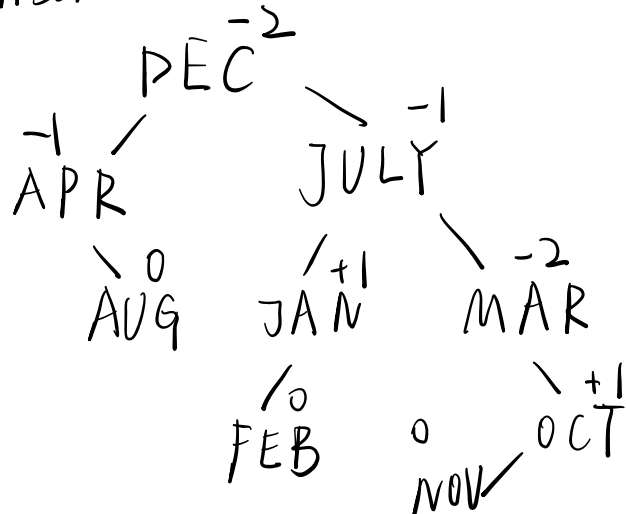
⑦ Insert OCT



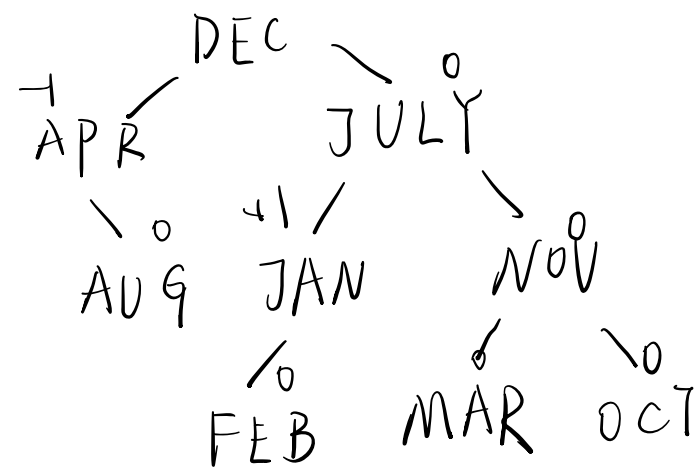
⑧ Insert FEB



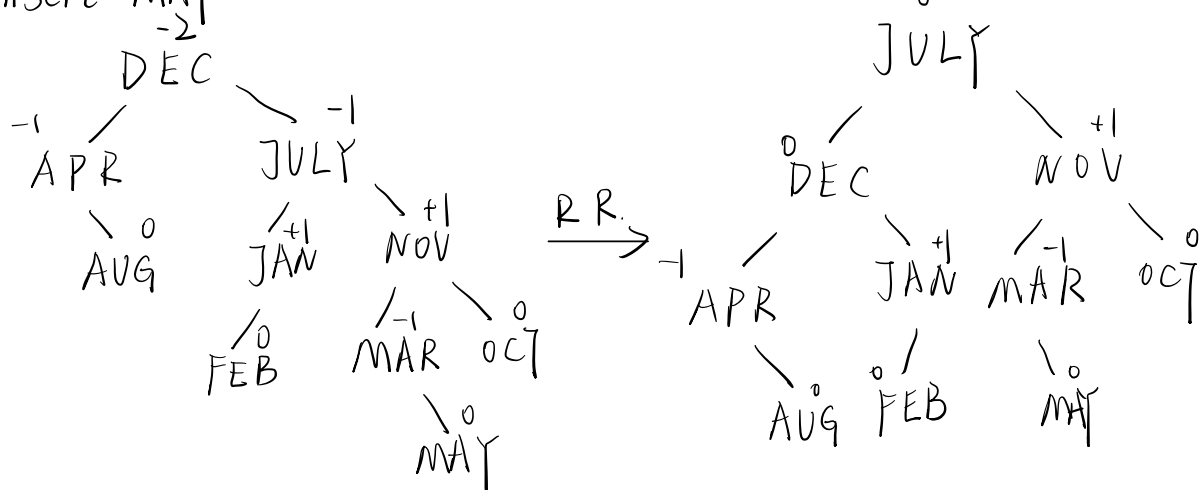
⑨ Insert NOV



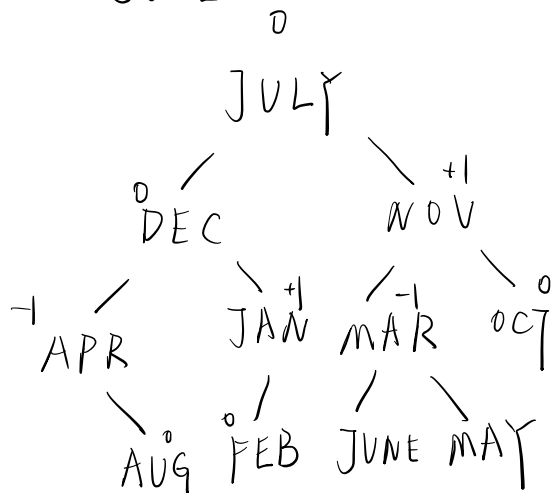
RL →



⑩ Insert MAY



⑪ Insert JUNE



多路查找树（基础概念辨析）

43. 下列叙述中，不符合m阶B树定义要求的是 (D)

- A. 根节点至多有m颗子树 ✓
- B. 所有的叶结点在同一层 ✓
- C. 各节点内关键字均升序或者降序 ✓
- D. 叶结点之间通过指针链接 ✗

44. 高度为5的3阶B树至少有2个结点，至多有12个结点。

$$\lceil m/2 \rceil = \lceil 3/2 \rceil = 2$$

$$\left. \begin{array}{l} ① \quad 1 \\ ② \quad 2 \\ ③ \quad 2^2 \\ \vdots \\ ⑤ \quad 2^{l-1} \end{array} \right\} 2^l - 1$$

$$\left. \begin{array}{l} ① \quad 1 = 3^0 \\ ② \quad 3 \\ ③ \quad 3^2 \\ \vdots \\ ⑤ \quad 3^{l-1} \end{array} \right\}$$

$$\frac{1(1 - 3^l)}{1 - 3} = \frac{3^l - 1}{2}$$

$$\begin{array}{l} 3^3 = 27 \\ 3^4 = 81 \\ 3^5 = 243 \end{array}$$

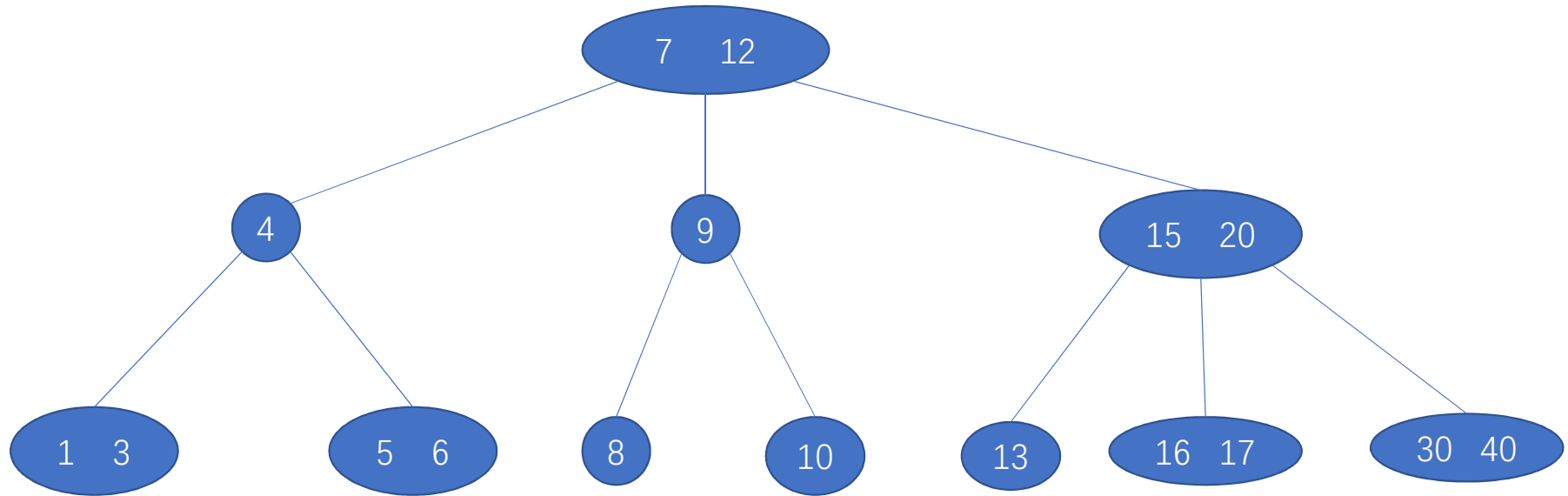
多路查找树（基础概念辨析）

45. [21年真题]B+树的优点0022(4)

- ①是m路搜索树✓
- ②可作为文件索引✓
- ③支持顺序索引✓
- ④支持随机索引✓

多路查找树（手撕B树）

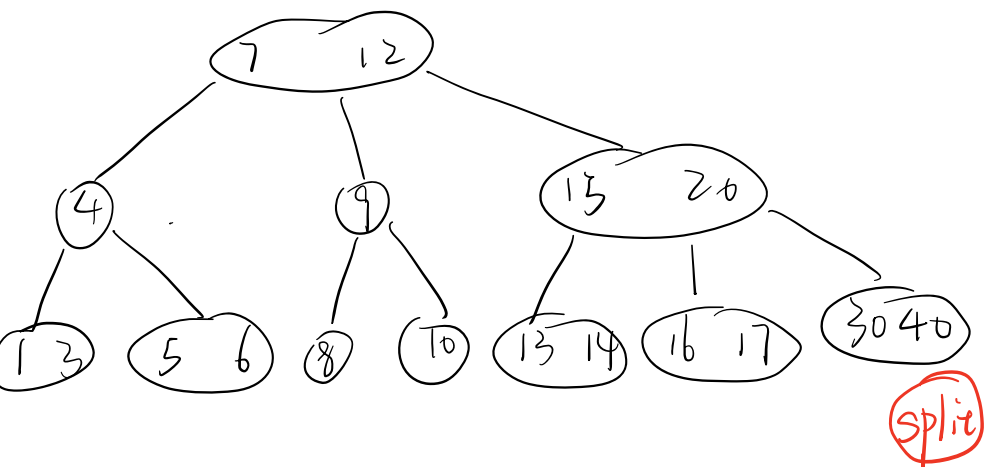
46. [from 吕建华老师PPT]对B树完成下列操作



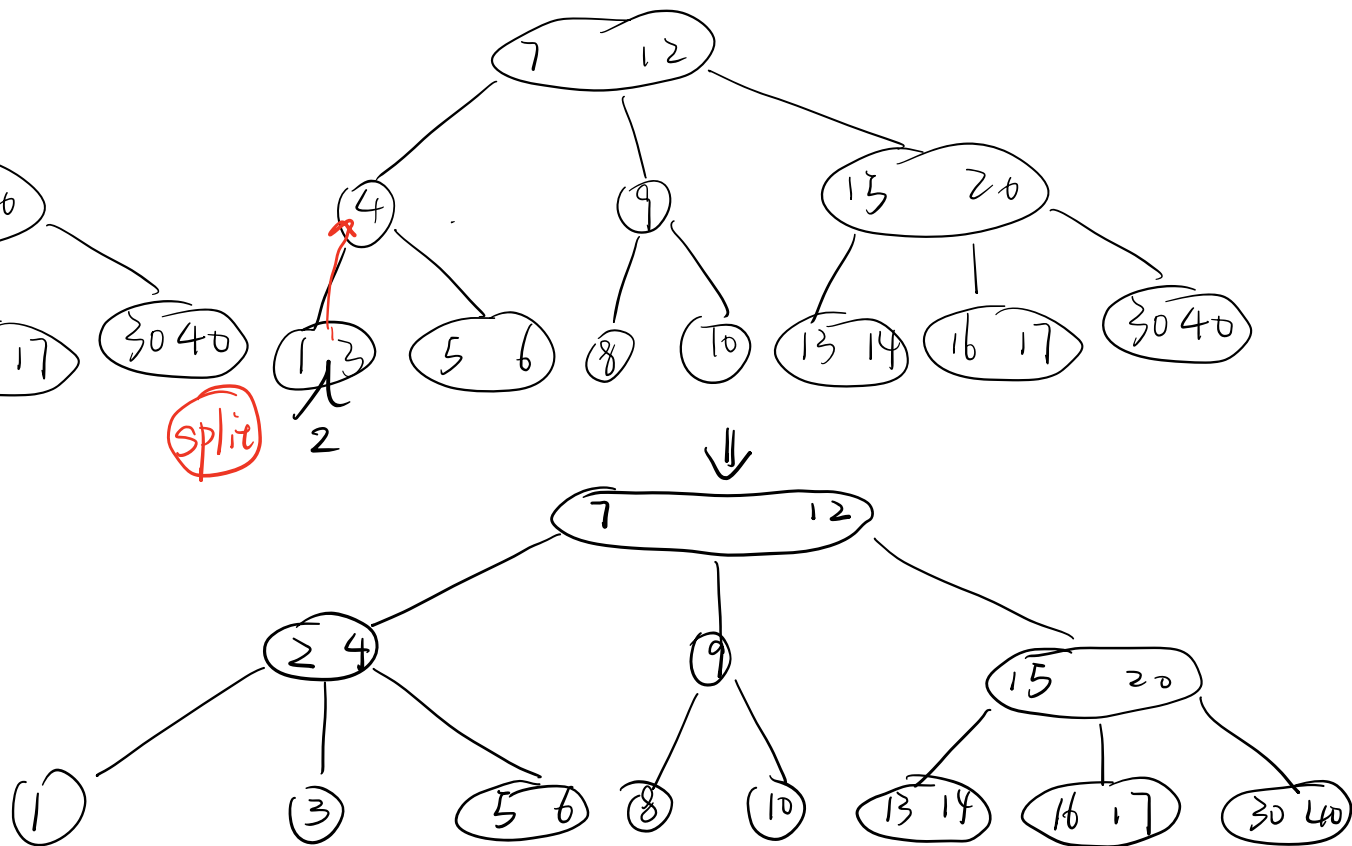
- ①Insert 14.
- ②Insert 2.
- ③Insert 18.
- ④Delete 7.
- ⑤Delete 3.
- ⑥Delete 8.

多路查找树（手撕B树）

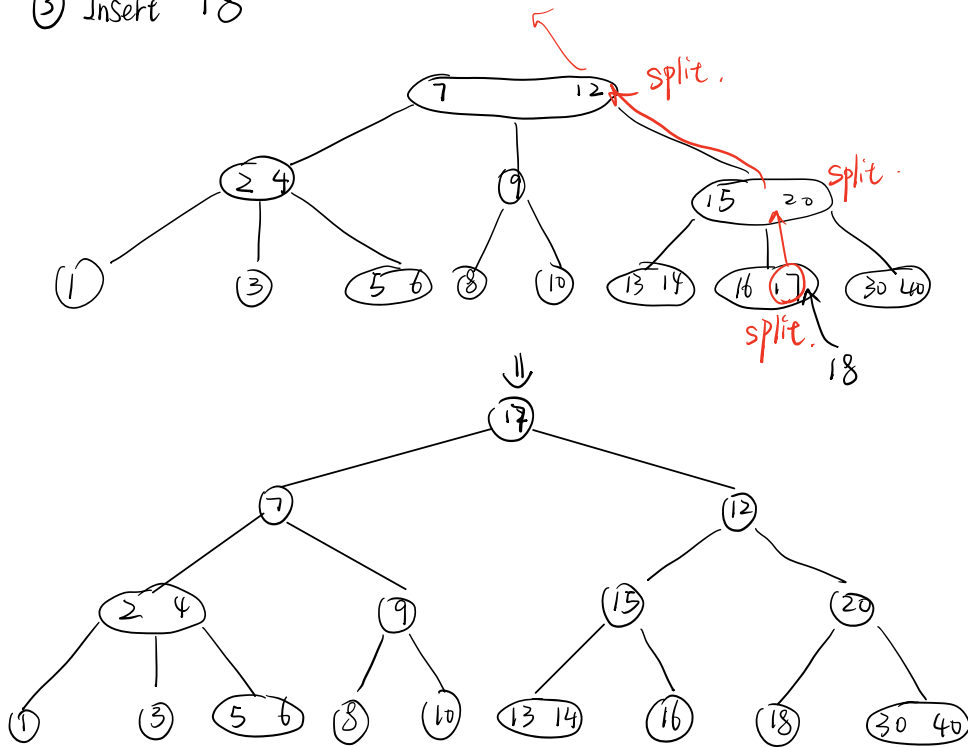
① Insert 14



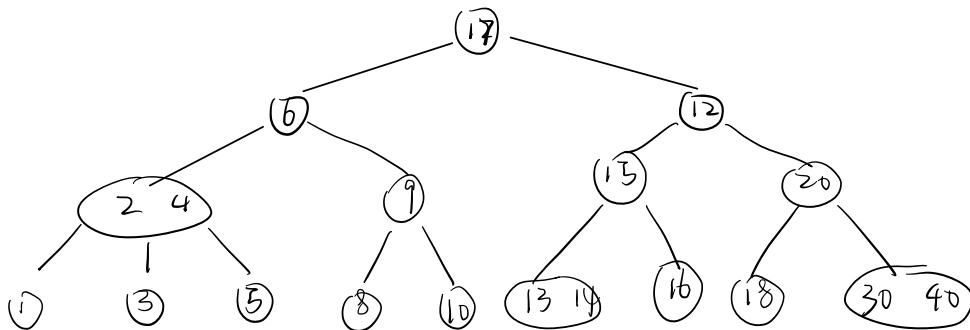
② Insert 2



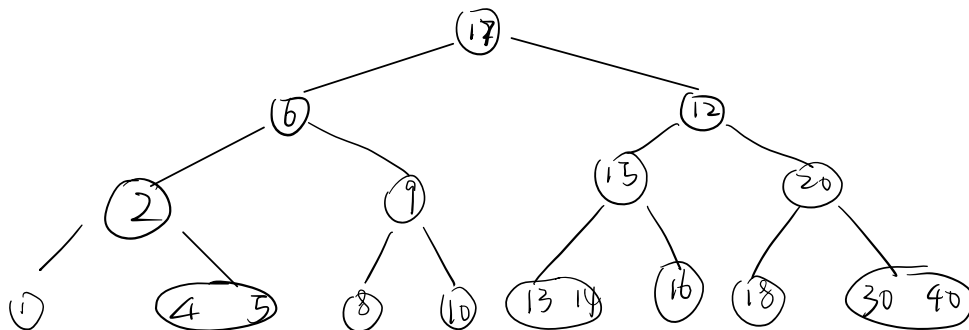
③ Insert 18



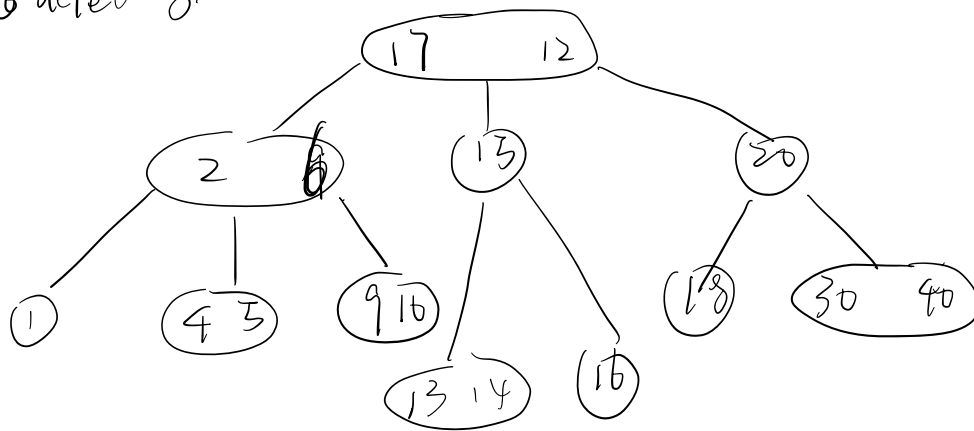
④ delete 7



⑤ delete 3



⑧ delete 8.



程序设计题举例1

输出指向单链表倒数第k个元素的指针（要求时间复杂度和空间复杂度均为 $O(n)$ ）

程序设计题举例2

判断一颗二叉树是否为二叉搜索树

程序设计题举例3

用邻接矩阵实现的图，实现算法判断结点u和结点v之间是否存在路径，并给出算法时间复杂度

谢谢大家！

「2022 青苗讲堂」

郑孝骥